

COMP 3311: Database Management Systems

Tutorial 4 Exercises

Relational Algebra (RA) and Structured Query Language (SQL)

Person(<u>personId</u> , personName, street, city)	Works(<u>personId</u> , <u>companyName</u> , salary)
Company(companyName, city)	Manages(personId, managerPersonId)

Exercise 1: Find persons who earn more than \$100,000 and live in Hong Kong. Include in the query result tuples attribute person name.

RA:

SQL: Complete your SQL statement in the **t4Exercises1-4.sql** script file.

Exercise 2: Find persons who either earn less than \$50,000 or more than \$100,000 and live in Hong Kong. Include in the query result tuples attributes person id and name.

RA:

SQL: Complete your SQL statement in the **t4Exercises1-4.sql** script file.

Exercise 3: Find persons who are not managers. Include in the query result tuples attributes person id and name.

RA:

SQL: Complete your SQL statement in the **t4Exercises1-4.sql** script file.

Exercise 4: Find persons who work for Alibaba and who live in the city where Alibaba is located. Include in the query result tuples attribute person name.

RA:

SQL: Complete your SQL statement in the **t4Exercises1-4.sql** script file.

Exercises 5, 6 and 7 are individual exercises.
EACH STUDENT MUST SUBMIT THEIR OWN SQL SCRIPT FILE.
A genuine effort must be made to complete all exercises to obtain full marks.
USE ONLY SQL CONSTRUCTS DISCUSSED IN THE COURSE.

Exercises 5, 6 and 7: Complete these exercises using **SQL Developer** by following the instructions given below.

1. **Download** the zipped folder **T4ExerciseFiles.zip** from the *Tutorial 4* entry of the *Tutorial Assignments* section on Canvas to the desktop and unzip it. The folder contains three SQL script files **employmentdb.sql**, **t4Exercises1-4.sql** and **t4Exercises5-7.sql**.
2. **Execute** the **employmentdb.sql** script file in **Oracle SQL Developer** to create the database against which you can run your exercise queries.
3. **Modify** the **t4Exercises.sql** script file by constructing an SQL statement for each of Exercises 5, 6 and 7 in the specified locations in the script file.
4. **Test** your queries to ensure that they execute without errors and get the expected result.

What To Submit

Your modified **t4Exercises5-7.sql** script file containing your SQL statements.

You must submit this completed script file. No other type of file submission is acceptable.

How To Submit

By 6:00 p.m. tomorrow, upload your modified **t4Exercises5-7.sql** script file to Canvas.

Note: You will only be able to submit a file with extension **sql**.

TEST YOUR SCRIPT FILE – USE ONLY SQL CONSTRUCTS DISCUSSED IN THE COURSE

The SQL queries in your submission will be tested by executing them against the **employmentdb** database. If any query raises an SQL error or if you use SQL constructs not discussed in the course, then you will receive at most 0.5 marks for this exercise. Therefore, test your SQL queries against the **employmentdb** database before you submit them and use only SQL constructs discussed in the course.